

storage oscilloscopes (DSOs) can store transient data for later study, analogue oscilloscopes have a larger dynamic range, and digital oscilloscopes offer higher bandwidth.

Most benchtop oscilloscopes are bulky. So, depending on the space restrictions in your lab or place of work, you can even consider some space-saving options like handheld oscilloscopes or those that are computer based.

Bandwidth

A conventional definition of an oscilloscope's bandwidth is the highest frequency it can measure without experiencing a considerable amount of distortion from the front-end amplifiers. Most oscilloscope manufacturers specify this at a 3dB level, which is equivalent to a 30% amplitude loss.

The entire oscilloscope, from probes and analogue inputs to CPU and all stages of signal processing, must be constructed to allow for a large bandwidth. This necessitates a significant investment in research and development as well as complex hardware design.

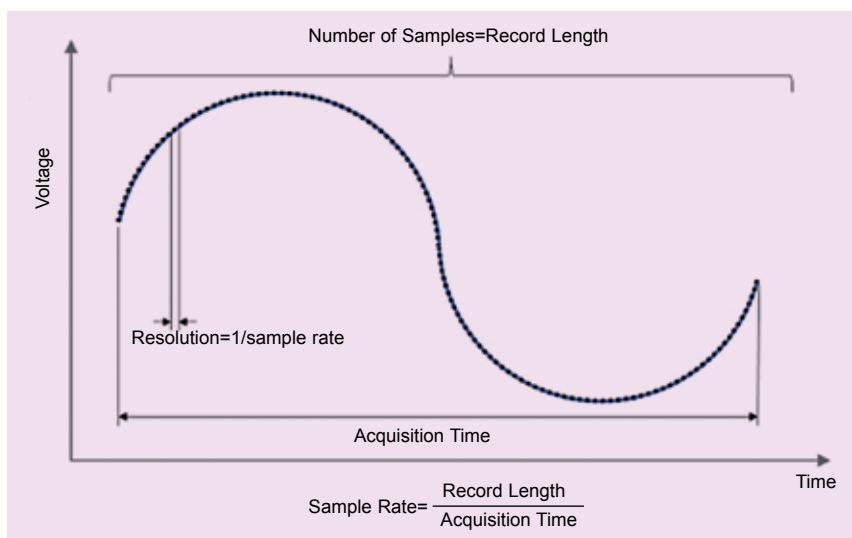
You will get a signal that is muted and warped if you choose an oscilloscope with insufficient bandwidth. Hence, make sure you select one whose bandwidth matches your circuit requirements.

Sample rate and memory depth

The number of samples (the record length) that an oscilloscope can

Various types of oscilloscopes

- Analogue Oscilloscope
- Digital Oscilloscope
- Digital Storage Oscilloscope (DSO)
- Digital Phosphor Oscilloscope (DPO)
- Mixed-signal Oscilloscope (MSO)
- Digital Sampling Oscilloscope
- Computer Based Oscilloscope
- Handheld Oscilloscope



Calculation of sample rate of an oscilloscope (Credit: www.oscilloscopesource.com)

collect each second is known as the sampling rate. The waveform's resolution is established by this. Because waveforms change quickly with time, you will not be able to see the signal clearly on the oscilloscope screen if its sample rate is too slow.

Nowadays, oscilloscopes have an automatic real-time sampling mode as the default sampling setting. However, most oscilloscopes let you set the sample rate yourself if you do not want the oscilloscope to choose it for you. But be sure to pick a sample rate that will offer sufficient detail to spot any unforeseen signal defects or oddities.

Another important thing to note is that the higher your sample rate is, the higher the memory requirement of your oscilloscope will be. This is because you need adequate memory to store the signal in order to examine and/or analyse it. Therefore, if you want a very fast sampling rate, make sure the oscilloscope has a good memory depth as well.

Instead of utilising the standard byte unit, manufacturers instead use the 'points' unit to describe the memory depth of their oscilloscopes. To ensure that the available memory is always filled, current

oscilloscopes automatically regulate the sample rate based on the time scale you have selected.

Rise time

An oscilloscope's capacity to recognise and record quickly rising and decreasing signals is indicated by its rise time. For instance, if the quickest rise time that an oscilloscope can detect is 3 nanoseconds, it means that it has a rise time of 3 nanoseconds. The oscilloscope must be capable of detecting quick changes in order to accurately display the waveform of such a signal.

If you choose an oscilloscope with a slow rise time, and you are working on a circuit that gives square wave outputs, then your waveform may not end up as sharp and square as you want it to be. So, when working with square waves, which have sharp edges, the rise time is especially crucial.

Probes and channels

Typically, oscilloscopes have two or four channels. You can work with more signals simultaneously if your oscilloscope has more channels. But a higher number of channels may lower the sampling rate.

Each channel has a unique connector where a probe can be at-